As lake vegetation and animal life decompose, their organic matter provides a source of nutrients. Lakes that have very high concentrations of nutrients (particularly nitrogen and phosphorus) are termed eutrophic and have low oxygen concentrations because of extensive microbial decomposition of organic matter. In comparison, the lakes that receive small amounts of nutrients and are nutrient sparse are oligotrophic. Caulobacter grows well in oligotrophic lakes.

The common microorganisms found in fresh water are *Pseudomonas, Flavobacterium, Serratia, Chromobacterium, Micrococcus, Aeromonas* and *Alcaligenes*.

As these waters reach the surface or inland bodies of water, they become contaminated with different types of microorganisms. Rivers flowing water (lotic) in close contact with the soil may contain large numbers of soil bacteria (Bacillus, *Actinomyces* and *Streptomyces*), Fungi (*Polyphagus, Penicillium* and *Aspergillus*), and algae (*Microcystis* and *Nostoc*). These microorganisms frequently impart an earthy odour and taste to river water. Rivers also receive high concentration of bacteria and agricultural chemicals through surface runoff water from adjoining soil during heavy rains and irrigation. In many countries, rivers are heavily polluted with sewage bacteria, especially *Escherichia coli, Enterobacter fecalis, Proteus vulgaris* and other intestinal bacteria.

Infobits

Red Tides

Red tide is a common name for a phenomenon known as an algal bloom which is caused by a species of Dinoflagellates (*Gymnodinium*) and the bloom takes on a red colour due to the presence of photosynthetic pigments with the production of toxins. The most conspicuous effects of red tides are associated with mortalities of marine species



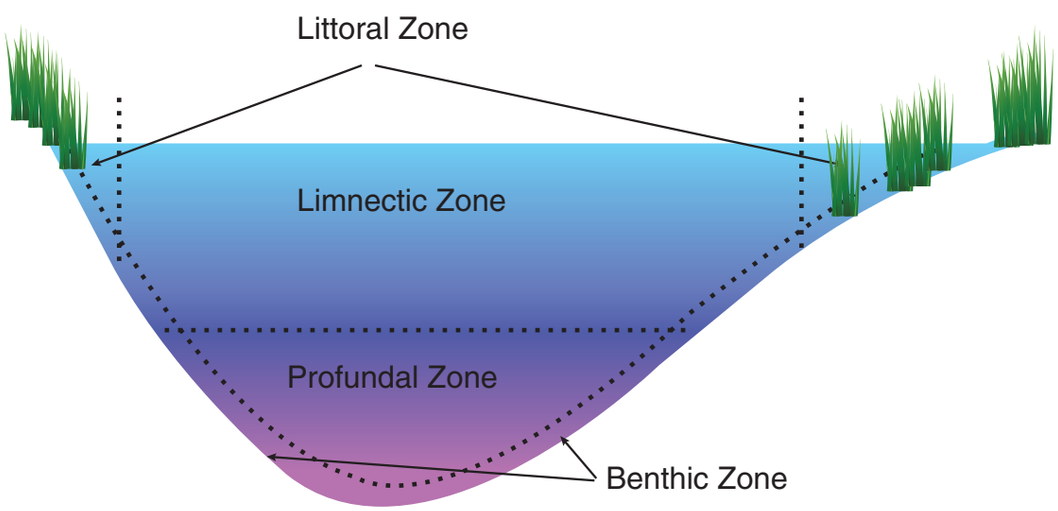


Figure 9.5: Light penetration zones of a freshwater lake

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